



OPEN Comparative study of the effects of different iron sources on bioavailability and gastrointestinal tolerability in iron-deficient rats

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Iron deficiency anaemia is a global health issue affecting millions of people. Though effective, conventional iron sources used for supplementation often cause gastrointestinal side effects. This preclinical study evaluated the efficacy and tolerability of three iron supplements—ferrous bisglycinate, LIPOFER™ microcapsules (Def-LFe1), and a commercially available microencapsulated iron pyrophosphate (Def-Fe2)—in reversing diet-induced iron deficiency in rats compared to FeSO₄. The study included three phases: (a) First, animals underwent a 24-day iron depletion period; (b) then, animals were exposed to a 21-day repletion period, receiving treatments at a human-equivalent dose of 80 mg of elemental iron; and (c) finally, to evaluate gastrointestinal tolerability, animals were supplemented with each treatment for 9 weeks. All supplements reversed iron deficiency within 14 days without adverse gastrointestinal effects. However, based on the decrease in TIBC and transferrin levels, coupled with the higher haemoglobin levels, Def-LFe1 demonstrated a higher absorption rate than Def-Fe2. In addition, the group supplemented with LIPOFER™ showed higher feed efficiency, especially compared to the control and FeSO₄ groups. FeSO₄ increased IL-6 gene expression in the colon, while Def-LFe1 did not. These findings highlight LIPOFER™ as a promising alternative to conventional iron supplementation. Further studies should be conducted to clarify these issues.

Keywords Iron supplementation, Ferrous sulfate, LIPOFER™, Iron bioavailability, Gastrointestinal side effects, Inflammation

In 2021, it was estimated that 1.9 billion people were affected by anaemia¹ primarily characterised by a deficiency in the number or quality of red blood cells, impairing the oxygen-carrying capacity of blood². Iron-deficiency anaemia (IDA) develops when there is not enough iron in the body due to increased iron demand, increased iron losses, and reduced absorption or intake of iron through the diet. IDA, the most common type of anaemia, can be a chronic condition which is often underdiagnosed due to its asymptomatic nature. When symptoms exist, they can present themselves as paleness of skin or conjunctiva. Other non-specific symptoms such as fatigue, weakness, and irritability can also be present, along with dizziness, shortness of breath and headaches. These symptoms collectively can have multiple consequences on human health, affecting overall quality of life and productivity.

Iron supplementation is the most common treatment for IDA. Iron salts such as ferrous sulfate (FeSO₄), gluconate, or fumarate, are the most prescribed sources of iron³. While these treatments are effective in increasing iron levels and alleviating IDA, they are often accompanied by side effects in the gastrointestinal tract, such as nausea, vomiting, constipation, and diarrhoea⁴. These side effects can lead to poor adherence to the treatment regimen, reducing its overall efficacy.

Given the challenges associated with iron salts such as FeSO₄ in terms of adverse effects on health, there is a pressing need for new strategies to prevent IDA. Chelation and microencapsulation are techniques used to increase iron bioavailability and reduce the associated side effects. Microencapsulation has gained considerable attention as a convenient delivery system for vitamins, minerals, and other active ingredients. Microencapsulation is the packaging of an active ingredient, such as iron, in a protective matrix. This protective matrix prevents the iron

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from interacting with other active ingredients, ensuring the stability of the iron is maximised. Microencapsulation can also attribute a specific release profile, resulting in iron release in the stomach or intestine, further optimising the absorption profile. Coupled with this, microencapsulation enhances the sensory aspect and increases the bioavailability of iron⁵. Additionally, microencapsulation improves its overall tolerability due to the ability to reduce interactions of non-absorbed iron with gastrointestinal mucosa and possibly gut microbiota^{6–8}.

In the present study, the capacity of three iron supplements—one chelated and two microencapsulated iron-based ingredients—to (a) recover from IDA in iron-deficient rats and (b) present the best tolerability profile as an alternative to FeSO₄ formulation was evaluated in a preclinical model of diet-induced Fe deficiency.

Materials and methods

Treatments

The products evaluated in the present study were the following iron supplements: Ferrous sulfate (FeSO₄) (Vidara, Spain), which contained 36.8% of elemental iron (368 mg/g); Ferrous bisglycinate (BisFe) (Vidara, Spain), which contained 20.3% of elemental iron (203 mg/g); microencapsulated iron pyrophosphate 1 (LIPOFER™ microcapsules) (Lipotec, S.A.U., Spain), which contained 9% of elemental iron (90 mg/g); microencapsulated iron pyrophosphate 2 (Sunactive™) (Taiyo GmbH, Germany), which contained 8.17% of elemental iron (81.7 mg/g). Animals were supplemented with the different treatments at the human extrapolated dose of 80 mg of elemental iron. The rat-equivalent dose (6.2 mg/kg bw of elemental iron) was calculated by applying the body surface area (BSA)-based conversion method described by Reagan-Shaw et al.⁹ using species-specific Km values derived from the respective body mass and length of the experimental animals (0.3 kg, 20 cm) and an established adult human reference (60 kg, 1.60 m)⁹.

Animals and experimental design

The Animal Ethics Committee of the UTNS-REUS (Reus, Spain) and the Generalitat de Catalunya approved all the procedures. The study was carried out following the Guide for the Care and Use of Laboratory Animals and in accordance with the European provisions of Directive 86/609/EEC on the protection of animals used for experimental and other scientific purposes. The study is detailed in accordance with ARRIVE guidelines and recommendations.

60 four-week-old male Wistar rats (ENVIGO, Barcelona, Spain) were used for this study. Rats were individually caged in the animal facility at 22 °C with a 12-h light/12-h dark cycle and were fed *ad libitum* a standard chow diet. Firstly, after 7 days of acclimation, rats were exposed to an iron depletion period. For that, all sixty animals were randomly distributed in 2 different groups: a Fe-deficient group (*n* = 50), which received a Fe-deficient diet (ENVIGO Teklad Custom Diet TD. 80396 containing 2–6 mg Fe/kg diet, which is a modification of the AIN-76 A diet) for 24 days to induce iron deficiency; and a control group (*n* = 10), which received a Fe-sufficient diet (ENVIGO Teklad Global 18% Protein Rodent Diet containing 200 mg Fe/kg diet). At the end of the depletion period, rats fed the Fe-deficient diet were assigned to 5 groups of 10 animals according to their haemoglobin (Hb) levels.

Then, rats were exposed to an iron repletion period, in which animals were supplemented with the different products of interest at the human extrapolated dose of 80 mg of elemental iron and fed the Fe-deficient diet for 21 days. All iron supplements were administered after a 5-hour fasting period to enhance its absorption. These treatments were dissolved in low-fat condensed milk and offered orally using a syringe, from which the rats voluntarily ingested the solution by licking. Finally, at the end of the repletion period, rats were exposed to both iron supplementation and a Fe-normal diet for 9 weeks to evaluate each formulation's gastrointestinal tolerability profile.

Thus, animals were distributed into six experimental groups: (1) The Control + Vehicle (Con) group was exposed to a Fe-normal diet during the depletion and repletion periods, as well as throughout the subsequent 9 weeks of treatment; this group received a placebo consisting of low-fat condensed milk. (2) The Fe-Deficient + Vehicle (Def-VH) group was subjected to the Fe-deficient diet during the depletion and repletion phases, followed by a Fe-normal diet for the 9-week treatment period, and also received a placebo (low-fat condensed milk). The remaining groups (3 to 6) followed the same dietary regimen as Def-VH group, but received different iron-based treatments dissolved in low-fat condensed milk: (3) The Fe-Deficient + Ferrous Sulfate (Def-FeS) group received ferrous Sulfate; (4) The Fe-Deficient + Ferrous Bisglycinate (Def-BisFe) group received ferrous bisglycinate; (5) the Fe-Deficient + Microencapsulated Iron (LIPOFER™) (Def-LFe1) group was administered microencapsulated iron (LIPOFER™); and, lastly, (6) the Fe-Def + Iron Microencapsulated 2 (Sunactive™) (Def-Fe2) received microencapsulated iron from Sunactive™.

The experimental design of the present study is represented in Fig. 1.

Analyses during the study

Body weight, food intake and feed efficiency

Body weight and food intake were recorded once each week during the three phases of the study. Feed efficiency (%) was calculated as the quotient between final body weight gain in grams and the total kilocalories consumed during the study and is expressed as a percentage of total caloric intake.

Haemoglobin parameters analysis

For the analysis of circulating levels of haemoglobin (g/dL), at the end of the depletion period (day 0) and after 14 and 21 days of iron supplementation, fresh whole blood samples were extracted from saphenous vein samples and treated with dipotassium ethylenediaminetetraacetic acid (EDTA) for the subsequent hematologic analysis. This analysis was performed using the Veterinary Haematology Analyzer Element HT-5 (HESKA, United States).

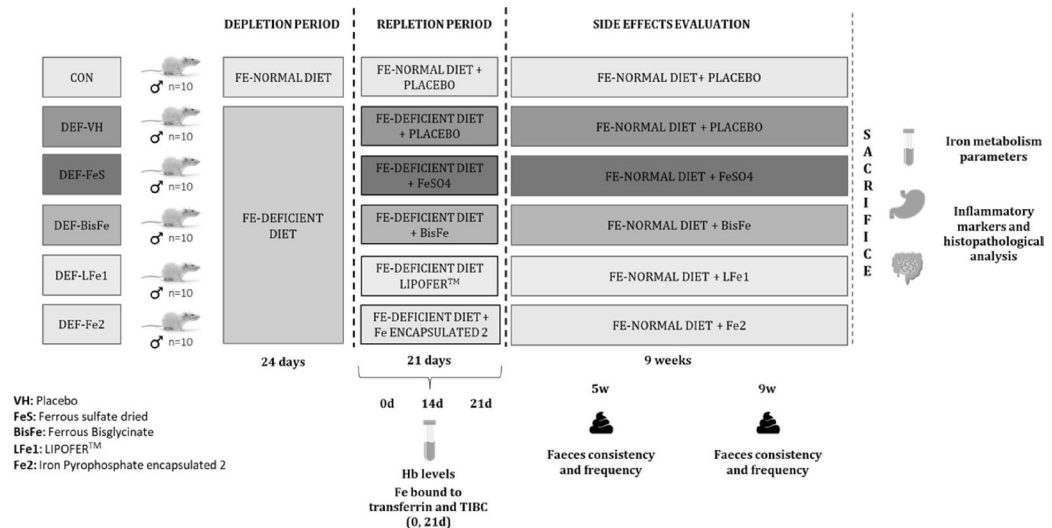


Fig. 1. Experimental design.

Iron content in Hb (in mg) was calculated by the Eq. (1), assuming that the total blood volume was 6.7% of the rat's body weight and an average Fe content of Hb was 0.335%¹⁰:

$$Hb - Fe (mg) = body\ weight\ (kg) \times \frac{0.067\ litres\ blood}{body\ weight\ (kg)} \times \frac{Hb\ (g)}{blood\ (L)} \times \frac{3.35\ mg\ Fe}{Hb\ (g)} \quad (1)$$

Values obtained from the iron content in Hb and Fe from food intakes were used to estimate the haemoglobin regeneration efficiency (HRE) according to the Eq. (2):

$$HRE = \frac{[mg\ Hb - Fe\ (repletion)] - [mg\ Hb - Fe\ (depletion)]}{Fe\ intake\ (mg)} \quad (2)$$

To assess the bioavailability of the different compounds, several bioavailability indices were calculated. The term bioavailability can be described as the “degree to which an ingested nutrient in a particular source is absorbed in a form that can be utilised in metabolism by the animal”. In many instances, it is essential to know the bioavailability of one substance relative to another. The bioavailability of a test substance relative to a reference substance is the ratio x_R/x_T , where x_R and x_T are amounts of the standard and test substances, respectively, that produce the same results. Formally, this ratio is the relative bioavailability value, $RBV = x_R/x_T$ ¹¹.

The bioavailability of Fe from FeSO₄ was considered as the reference and was therefore assigned a value of 1. Hence, the relative Fe bioavailability, or relative biological value (RBV) based on HRE (RBV_{HRE}) was calculated by dividing the individual HRE values of the Fe source by the mean HRE values of FeSO₄ at a given dose (Eq. 3).

$$RBV_{HRE} = \frac{HRE \times group}{HRE\ FeSO_4\ group} \quad (3)$$

The Hb change (ΔHb) was calculated for each animal by subtracting the Hb repletion and depletion values. The relative biological values based on this change ($RBV_{\Delta Hb}$) was calculated for each animal according to the Eq. (4):

$$RBV_{\Delta Hb} = \frac{\Delta Hb \times group}{\Delta Hb\ FeSO_4\ group} \quad (4)$$

Since the Hb concentration of rats fed the Fe-def diet continued to decrease during the repletion period, the mean ΔHb for the Def-VH group was subtracted from the respective ΔHb (Eq. 5).

$$Corrected\ \Delta Hb = \Delta Hb\ (each\ animal) - \Delta Hb\ (average\ for\ Def - VH\ group) \quad (5)$$

Iron parameters

For the analyses of iron levels (Fe) (µg/dL) and total iron binding capacity (TIBC) (µg/dL) at days 0 and 21, serum was obtained by centrifugation and stored at - 80 °C until analysis. These parameters were quantified using colourimetric commercial kits for iron (QCA, 991342) and TIBC (QCA, 991462).

TIBC indicated the maximum amount of iron needed to saturate plasma or serum transferrin, which is the primary iron-transport protein. Therefore, the transferrin saturation (TS) was calculated according to the Eq. (6):

$$\text{Transferrin saturation (TS)} = \text{Serum Fe/TIBC} \times 100 \quad (6)$$

Theoretically, 1 mg of transferrin can bind a maximum of 1.4 µg of iron¹². Therefore, TIBC correlates well with transferrin concentration and was estimated according to the Eq. (7):

$$\text{Transferrin (mg/dL)} = \text{TIBC (µg/dL)} / 1.4 \quad (7)$$

Since TIBC measures the blood's capacity to bind iron with transferrin, the unsaturated iron binding capacity (UIBC) was obtained as follows (Eq. 8):

$$\text{UIBC} = \text{TIBC} - \text{Serum Fe} \quad (8)$$

Faeces consistency and water content

To evaluate the consistency and water content, in the 5th and 9th week, fresh faeces were collected during a 2-hour interval from 8:00 to 10:00 h. At the end of this period, the number of stools was recorded and weighed. The qualitative appearance (consistency) was determined visually following a 3-point scale: normal (3 points), semi-solid (2 points) and diarrhoea (1 point) according to the procedure described by Sung et al.¹³. The faeces were then dried in an oven for 24 h at 70°C and weighed again. The water content was determined by calculating the difference between the weight of fresh and dry faeces, expressing the result in %: (water content (mg) / fresh faeces (mg)) * 100.

Weight, number, and frequency of bowel movements evaluation

Faeces deposited during the next 22 h were collected. Their weight, number and frequency of bowel movements were determined from the quantification of the number of faeces made by each animal during the first 2 h and the following 22 h, as well as the record of its total fresh and dry weight (after being dried in an oven for 24 h at 70 °C).

Analyses at the endpoint

After 9 weeks, rats were sacrificed under anaesthesia (pentobarbital sodium, 60 mg kg⁻¹ bw, Merck, Danvers, MA, USA) after 6 h of diurnal fasting, administered intraperitoneally. Euthanasia was completed by exsanguination via cardiac puncture. Serum and plasma were obtained by centrifugation and stored at -80 °C until analysis. Stomach, small and large intestines were rapidly removed, weighed, frozen in liquid nitrogen and stored at -80 °C until analysis. The liver and cecum were kept for possible analyses of interest to the company.

Iron-related parameters analysis

At the end of the study, a portion of blood was processed to obtain serum, and another portion was treated with dipotassium ethylenediaminetetraacetic acid (EDTA) to analyse haemoglobin levels in fresh blood. Iron bound to transferrin levels and TIBC were measured in serum.

Hepcidin analysis

Circulating hepcidin levels were measured by an ELISA kit (Abcam, Cambridge, United Kingdom) at the beginning of the repletion period and at the end of the study to calculate the hepcidin induction ratio.

Cytokines analysis

Twenty and fifty milligrams of duodenum and stomach, respectively, were homogenised in one mL of tissue lysate buffer and centrifuged at 10,000 x g for 5 min at 4°C. Protein quantification of the homogenates was performed by BCA protein assay kit (ThermoFisher, Massachusetts, United States). Levels of the pro-inflammatory chemokine TNF-α and anti-inflammatory chemokine IL-10 were analysed in the stomach and duodenum using an ELISA kit (ThermoFisher, Massachusetts, United States) as markers of inflammation.

Gene expression analysis

The total RNA was extracted from 20 mg of duodenum and colon samples from Def-VH, Def-FeS and Def-LFeI groups, using Trizol Reagent (Thermo Fisher Scientific, Barcelona, Spain) according to the manufacturer's protocol. The RNA yield was quantified on a NanoDrop 1000 spectrophotometer (Thermo Scientific, Wilmington, DE, USA), measuring the sample absorbance at 260 nm and 280 nm and using the 260/280 ratio to assess RNA purity. RNA purity was considered adequate when the 260/280 ratio was ≥ 1.9. To analyse the expression of the samples, cDNA was synthesised using the High-Capacity cDNA Reverse Transcription Kit (Applied Biosystems, Barcelona, Spain). A MyGene L Series Peltier Thermal Cycler (LongGene, Hangzhou, China) was used for reverse transcription. The reaction was performed according to the manufacturer's instructions. The cDNA was subjected to a quantitative reverse transcriptase polymerase chain reaction amplification using LightCycler 480 SYBR green I Master-20 (Roche Diagnostics, Barcelona, Spain) in a Light Cycler 480II (Roche Diagnostics, Barcelona, Spain). Specifically, the expression of zonulin-1 (ZO1), occludin (OCLN), claudin-3 (CLDN-3) and mucin-2 (MUC-2) was evaluated, representing genes encoding proteins involved in intestinal permeability. Finally, the expression of the pro-inflammatory interleukins IL-1β and IL-6 was assessed. The primers used for the different genes were obtained from Biomers.net (Ulm, Germany) and the sequences were as follows: ZO1, forward 5'-GACCGATTGCTGATGTTGCC-3' and reverse 5'-TGAAGGCGAATAATGCCCGA-3'; OCLN, forward 5'-TGAACAGCCCCCTAATGTGG-3' and reverse 5'-TGCCATTCACCTTTGCCGTTG-3'; CLDN3, forward 5'-ACCGCACCATCACCCTAC-3' and reverse 5'-CGTGGCGTCTGTAAACCATC-3'; MUC2, forward 5'-AGGTTCTCGGGAATGTTGC-3' and reverse 5'-ACCTGTCGTTGGGGTTTTTCT-3'; IL1β, forward 5'-AAATGCCTCG

TGCTGTCTGA-3' and reverse 5'-AGGCCACAGGGATTTTGTCTG-3'; *Il6*, forward 5'-GTCAACTCCATCTG CCCTTCAG-3' and reverse 5'-GGCAGTGGCTGTCAACAACAT-3'; *Ppia*, forward 5'-CCAAACACAAATGG TTCCAGT-3' and reverse 5'-ATTCCTGGACCCAAAACGCT-3'. The fold changes in the mRNA levels were calculated using the $-2\Delta\Delta C_t$ method¹⁴ with the PPIA gene as an endogenous control.

Histological analysis

For histology, tissue samples measuring approximately 2 cm were excised from the stomach and duodenum post-mortem; they were fixed in 10% formalin, dehydrated, and embedded in paraffin. Sections measuring 5- μ m were cut longitudinally at the midline of the paraffin-embedded duodenal samples. For the stomach, sections were stained with haematoxylin/eosin, whereas for the duodenum, one section was stained with haematoxylin/eosin and another section with a Prussian blue iron staining. The number of iron foci detected in duodenal samples was quantified.

Statistical analysis

Statistical analyses were performed using the IBM SPSS Statistics 26.0 program. Grubbs' test was used to detect outliers, which were discarded for subsequent analyses. The normality assumption was determined using the Kolmogorov-Smirnov test, and the homoscedasticity among groups was assessed using Levene's test. When one or both conditions were unmet, data was transformed to a base-10 logarithm to obtain a normal distribution.

For the parameters measured on several occasions, the ANOVA test of repeated measures (rm-ANOVA) was used (t , time effect; T , treatment effect; txT , time x treatment interaction effect; $p < 0.05$). Differences among the treatments were evaluated by the one-way ANOVA (1w ANOVA) test followed by Duncan's new multiple range test (as a post hoc test). In contrast, differences between diets were evaluated by the Mann-Whitney test, as indicated in the respective figure legend. Student's t -test was used for single statistical comparisons. Data are presented as the mean $\pm$ standard error of the mean (SEM). Statistical significance level was set at bilateral 5% ($p < 0.05$).

Results

Body weight, food intake and feed efficiency

No treatment effects were reported among groups regarding weekly and final cumulative food intake (Fig. 2C and D). Body weight and body weight gain were reported weekly (Fig. 2A and B). During both repletion and side effects evaluation periods, all groups displayed a gradual increase in their body weight (t effect, $p < 0.05$, rm-ANOVA) (Fig. 2A). Interestingly, in weeks 7, 9, 10, 11, 13, 14, and 17, Def-LFe1 animals exhibited significantly higher body weight gain than those supplemented with $FeSO_4$ (Fig. 2A). Def-LFe1 animals also resulted in the highest mean body weight gain. No significant differences were reported among the other Fe-supplemented groups. Figure 2E illustrates that animals supplemented with LFe1 exhibited all groups' highest mean feed efficiency values. The feed efficiency values for animals supplemented with LFe1 were significantly higher than the Con and Def-FeS groups. This indicates an increased conversion of energy intake into body weight for the Def-LFe1 group of animals.

Haemoglobin parameters

After an iron depletion period of 24 days, a significant decrease in blood haemoglobin levels was reported in iron-deficient diet-fed animals (12.7 ± 0.2 g/dL) in comparison with control diet-fed animals (18.4 ± 0.3 g/dL) ($p < 0.001$ Mann-Whitney test) (Fig. 3A).

To assess the capacity of the different iron supplements to reverse iron deficiency, haemoglobin levels were monitored at 14 and 21 days of the repletion period. As shown in Fig. 3B–D, all the iron supplement-treated groups significantly increased haemoglobin levels at days 14 and 21 in comparison to day 0, showing a complete recovery of haemoglobin levels (t , $p < 0.001$, rm-ANOVA) (Figs. 3B–D). However, no significant differences among iron-treated animals were reported on days 14 and 21 (Figs. 3B–D).

The HRE was determined at days 14 and 21 of the repletion period in all the supplemented groups. As shown in Fig. 4, the best repletion efficiency was reported on day 14 of iron supplementation, while a significant decrease in HRE levels was observed on 21 days in all the groups (Fig. 4A). No differences were reported among Fe-supplemented groups.

Additionally, for the analysis of the relative bioavailability value (RBV) after the repletion period, Def-FeS animals were used as the reference group (RBV_{HRE} and corrected $RBV_{\Delta Hb}$ of De-FeS group = 1).

As shown in Fig. 5, the three iron supplements tested in the present study exhibited a similar iron bioavailability to the reference compound ($FeSO_4$) at days 14 and 21, with no significant differences reported among the different groups.

Iron metabolism-related parameters

After 21 days of iron supplementation, iron metabolism-related parameters significantly changed between animals fed an iron-deficient diet and their counterparts fed a control diet (Fig. 6). Specifically, lower levels of serum iron (Fig. 6A) and transferrin saturation (Fig. 6B) and higher levels of transferrin (Fig. 6C), UIBC (Fig. 6D) and TIBC (Fig. 6E) were reported in deficient diet-fed animals when compared to control diet-fed ones.

As shown in Fig. 7, after 21 days of iron repletion, all iron-supplemented groups displayed a significant increase in serum iron levels and transferrin saturation and a sharp decrease in TIBC, transferrin and UIBC levels compared to Def-VH animals.

Although, overall, no significant changes among groups were found under the one-way ANOVA, a Student's t -test carried out between LFe1 and Fe2 groups revealed a residually higher decrease in TIBC ($p = 0.047$, Student's

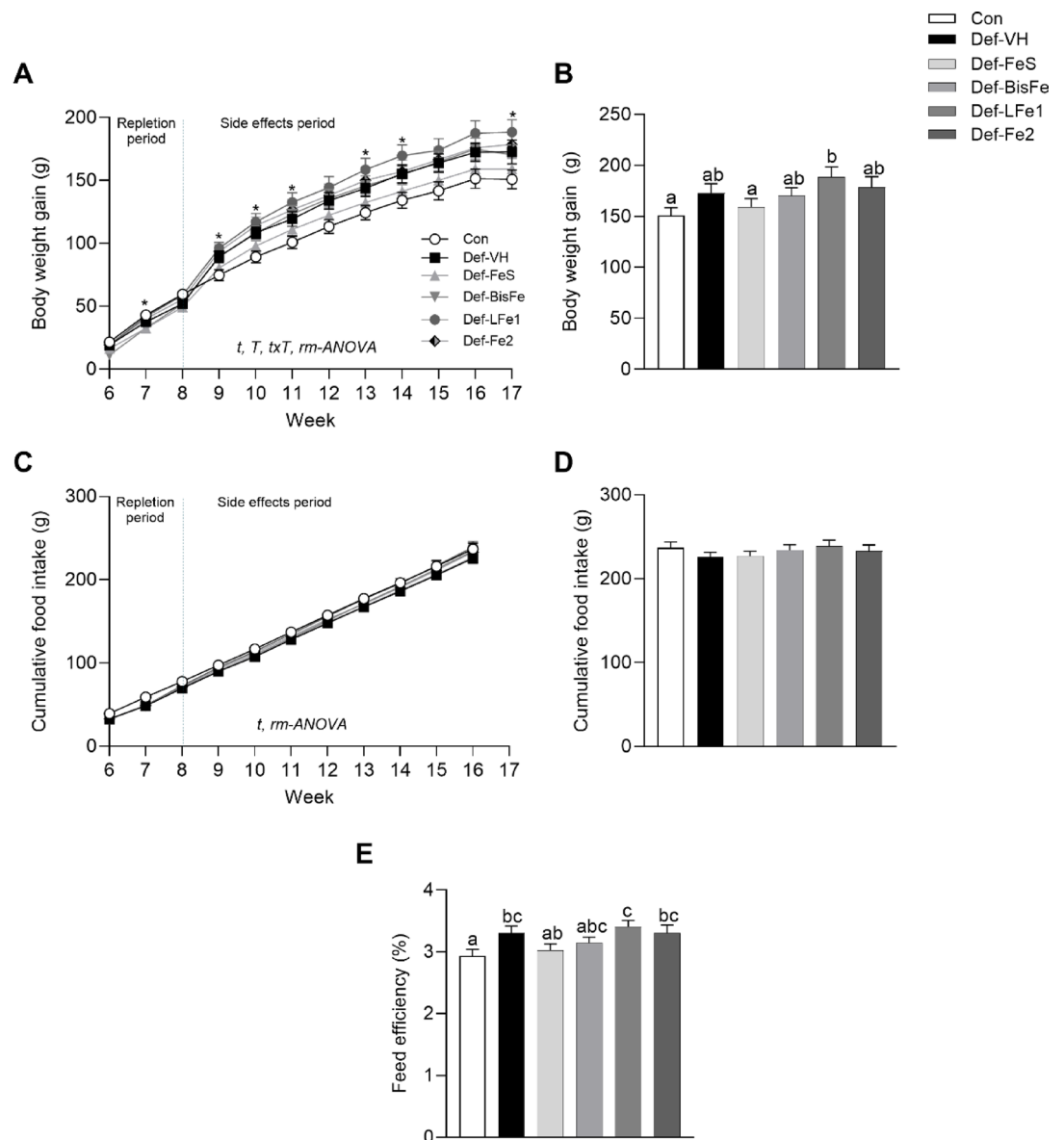


Fig. 2. Weekly body weight gain (A) and cumulative food intake (C), body weight gain (B), cumulative food intake (D) and feed efficiency (%) (E) at the end of the study. All values are presented as mean $\pm$ SEM. *t*, time effect; *T*, treatment effect; *txT*, time $\times$ treatment effect ($p < 0.05$, rm-ANOVA). * Def-FeS vs. Def-LFe1 ($p < 0.05$, 1w-ANOVA, Duncan's post hoc). ^{abc} Mean values with different letters are significantly different among the groups ($p < 0.05$, 1w-ANOVA, Duncan's post hoc test).

t-test) (Fig. 7F) and transferrin levels ($p = 0.047$, Student's *t*-test) (Fig. 7I) in those animals supplemented with LFe1.

At the end of the study, iron-related parameters were evaluated to assess iron metabolism after 9 weeks of daily iron supplementation (Fig. 8). Relevantly, the Def-LFe1 group exhibited significantly higher haemoglobin levels than the Def-Fe2 group and their counterparts supplemented with the other products, except Def-BisFe animals (Fig. 8A). This difference was not reported in serum Fe and TIBC levels (Fig. 8B and C). In addition, as noted at the end of the repletion period, the Def-LFe1 group exhibited significantly lower TIBC levels than Def-Fe2 animals (Fig. 8C).

Bowel movement assessment

Firstly, faecal water content was quantified in the different experimental groups at weeks 5 and 9 to identify possible adverse effects of each iron supplement (Fig. 9A and B). No differences were reported in both week 5 and week 9 among iron-treated groups. Similarly, the evaluation of weight, number and frequency of faecal depositions revealed no significant differences among the experimental groups (Fig. 9C–H).

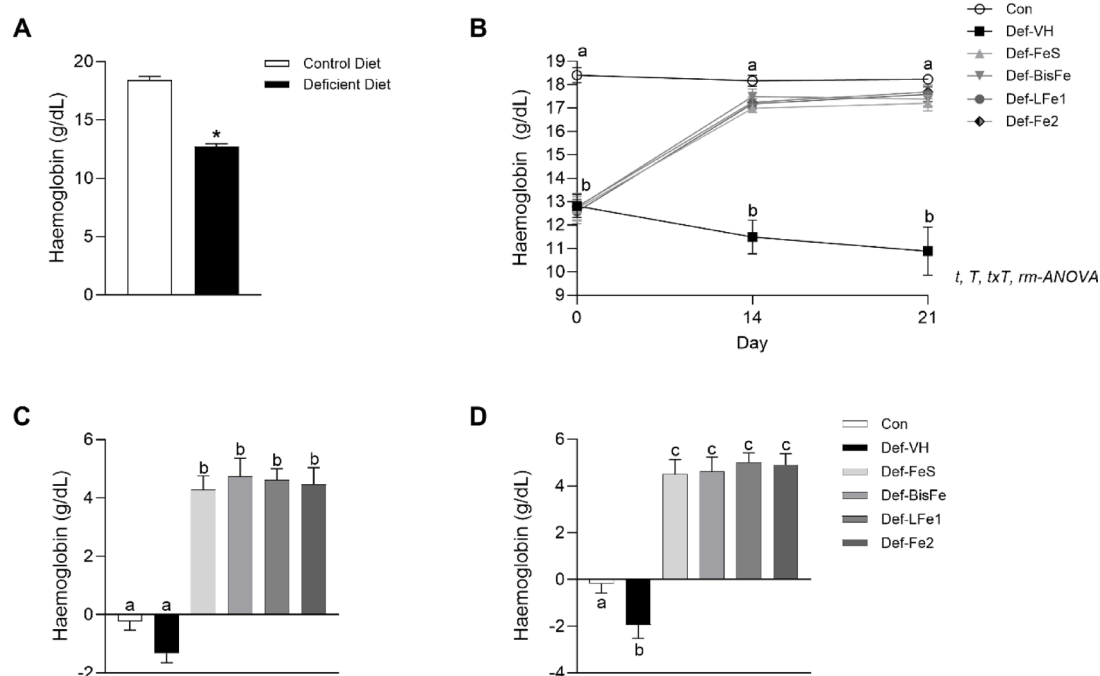


Fig. 3. Haemoglobin levels after 24 days of iron depletion period in control and deficient diet-fed animals (A), haemoglobin changes (B) after 14 (C) and 21 (D) days of iron repletion. All values are presented as mean $\pm$ SEM. * $p < 0.001$ Mann-Whitney test vs. Con group. ^{abc} Mean values with different letters are significantly different among the groups ($p < 0.05$, 1w-ANOVA, Duncan's post hoc test). *t*, time effect; *T*, treatment effect; *txT*, time x treatment interaction effect, $p < 0.05$, Repeated measures ANOVA (rm-ANOVA).

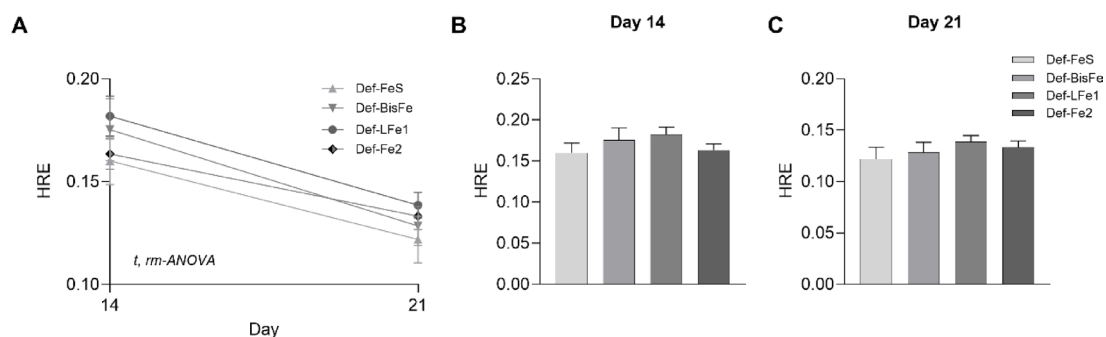


Fig. 4. HRE (A) in Fe-depleted rats after 14 (B) and 21 (C) days of repletion period. All values are presented as mean $\pm$ SEM. *t*, time effect ($p < 0.05$, rm-ANOVA).

Biometrical parameters

No differences were reported in liver, stomach, large intestine and colon weight among the experimental groups (Fig. 10A, B, C and E). Similarly, no differences in large intestine and colon length were reported among groups (Fig. 10D and F). Finally, the Def-Fe2 group exhibited a significantly higher cecum weight than FeS-treated animals (Fig. 10G).

Hepcidin analysis

At the end of the iron depletion period, the plasmatic hepcidin levels measurement showed significantly lower hepcidin levels in those animals fed an iron-deficient diet compared to their control-fed diet counterparts (Fig. 11A).

In the present study, the hepcidin induction ratio (Fig. 11C) was calculated from the plasmatic hepcidin levels measured at the beginning of the repletion period and those obtained at the end of the study (Fig. 11B). All the experimental groups of the study showed an increase in hepcidin levels at the end of the study, evidenced by the ratio higher than 1 displayed by all the groups (Fig. 11C). However, no differences were reported among the Fe-treated groups.

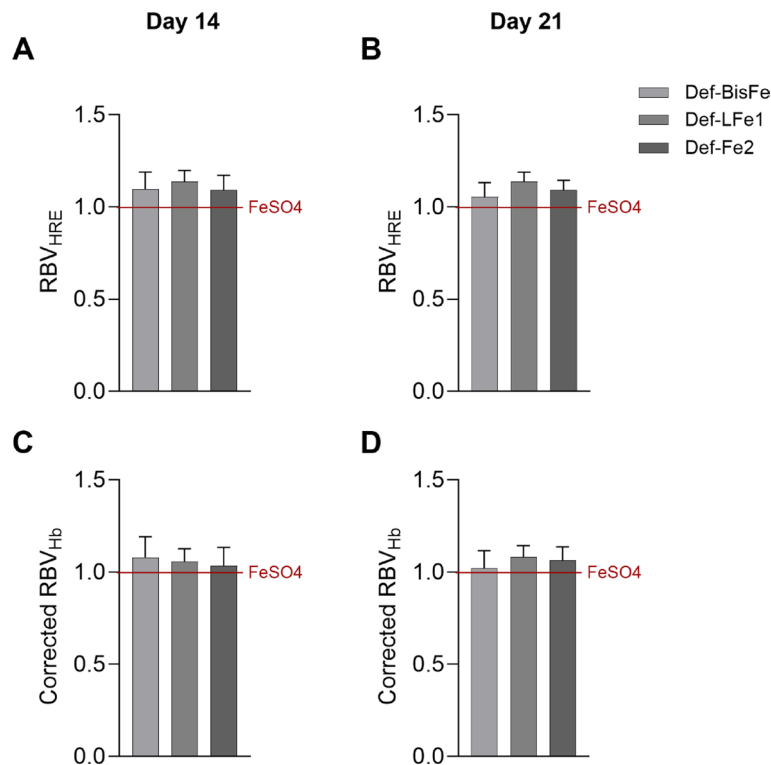


Fig. 5. RBV_{HRE} (A, B) and corrected RBV_{Hb} (C, D) in Fe-depleted rats after 14 and 21 days of repletion period. All values are presented as mean $\pm$ SEM.

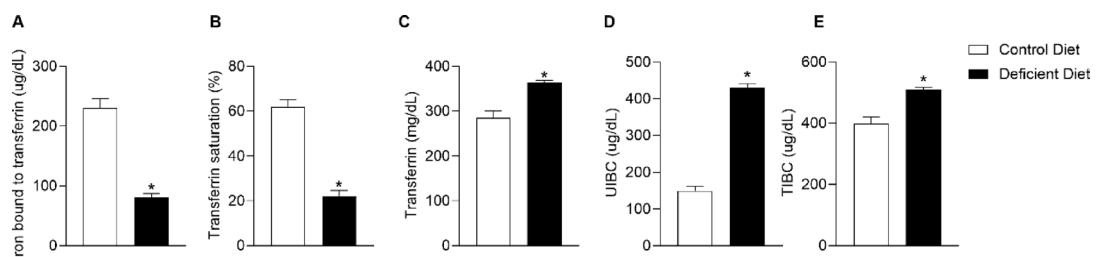


Fig. 6. Iron metabolism-related parameters after 24 days of iron depletion in control ($n = 10$) and deficient diets-fed animals ($n = 50$). All values are presented as mean $\pm$ SEM. * $p < 0.001$ vs. Con group, Mann-Whitney test.

Cytokines analysis

No significant differences were found among iron-supplemented groups in both stomach and duodenal samples regarding TNF α and IL10 levels (Fig. 12).

Gene expression analysis

In the duodenum, a decrease in *OCN* mRNA levels was reported in those animals supplemented with Def-LFe1 compared to Def-VH animals (Fig. 13A). In addition, a numerical reduction in *ZO1* and *MUC2* was also reported in both groups supplemented with iron, in comparison with animals supplemented with the placebo. However, these differences were not statistically significant (Fig. 13A).

In colon samples, Def-FeS animals displayed a trend to decrease *MUC2* mRNA levels compared to Def-LFe1 animals ($p = 0.056$, Student's *t*-test) (Fig. 13B). In addition, in terms of inflammation, Def-FeS animals exhibited a significant increase in *IL6* mRNA levels compared to Def-LFe1 animals (Fig. 13B).

Histological analysis

As illustrated in Fig. 14, no significant differences under the one-way ANOVA analysis were reported among the experimental groups in the study of villus height and width, crypt depth and the villus height to crypt depth ratio. As an exploratory analysis, pairwise comparisons between different experimental groups showed

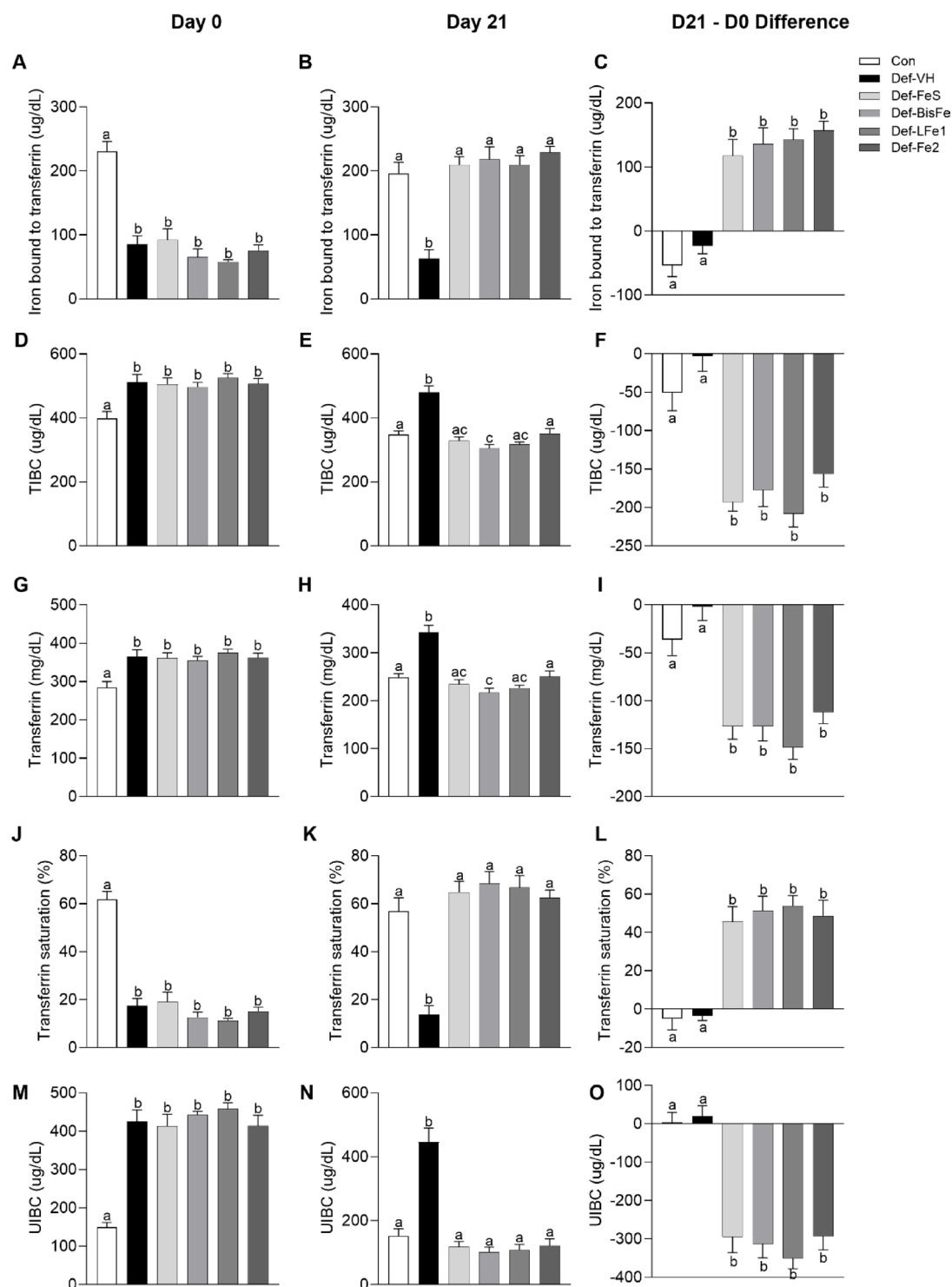


Fig. 7. Iron metabolism-related parameters at days 0 (basal levels after the depletion period), 21 (end of the depletion period) and D21-D0 variation of each parameter. All values are presented as mean $\pm$ SEM. ^{abc} Mean values with different letters are significantly different among the groups ($p < 0.05$, 1w-ANOVA, Duncan's post hoc test).

a residually significant higher villus height to crypt depth ratio in Def-LFe1 and Def-Fe2 groups ($p = 0.025$ and $p = 0.044$, Student's t-test, respectively) in comparison with Def-FeS animals (Fig. 14D).

In addition to the histological analyses, iron foci in the duodenal tissue were quantified in the different experimental groups.

As shown in Fig. 15, despite the remarkable variability observed in the duodenal sections across different animals, a clear difference was observed between iron-treated and control groups treated with placebo. Focusing

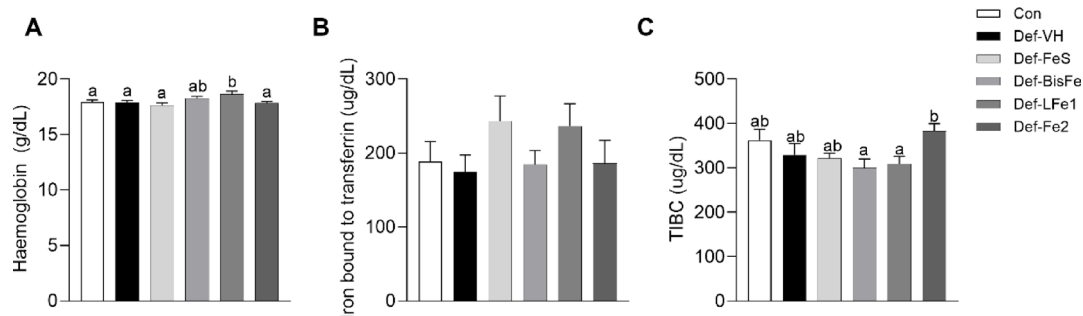


Fig. 8. Haemoglobin (A), iron bound to transferrin (B) and TIBC (C) levels at the end of the study. All values are presented as mean $\pm$ SEM. ^{abc} Mean values with different letters are significantly different among the groups ($p < 0.05$, 1w-ANOVA, Duncan's post hoc test).

on Fe-supplemented animals, Def-BisFe animals displayed significantly lower iron foci in gut lamina propria than their counterparts. Def-LFe1 animals displayed a numerically higher amount of iron foci than Def-FeS and Def-Fe2 groups, although these results were not statistically significant (Fig. 15G).

Discussion

In the present study, all the tested iron supplements were shown to fully reverse iron deficiency after 14 days of supplementation. However, for the FeSO_4 -treated animals, several parameters, such as significantly lower body weight gain, lower feed efficiency, duodenal villus height to crypt depth ratio, and higher colonic IL6 gene expression, were reported. These would suggest potential side effects associated with FeSO_4 supplementation, including reduced nutrient absorption. Surprisingly, these were not reported in Def-LFe1 animals, highlighting the ability of LIPOFER™ to revert iron deficiency while potentially minimising iron-related side effects.

Food intake, body weight, and feed efficiency levels were monitored throughout the chronic iron supplementation to detect potential side effects. As mentioned, no differences were observed in food intake levels for each group, which indicated that calorie intake was also identical. In terms of body weight gain, as expected, the Con group's body weight gain was the lowest, most likely due to the difference in the calorific content of the diet versus the Fe-deficient groups (3.1 kcal/g vs. 4 kcal/g, respectively). The increased conversion of energy intake into body weight, i.e., feed efficiency demonstrated by Def-LFe1-treated groups in comparison with the Def-FeS group, explains, at least in part, the higher body weight gain reported in Def-LFe1 group compared to their Def-FeS counterparts. These results highlight the ability of LIPOFER™ supplementation to enhance feed efficiency and promote growth in this animal model. Previous studies have shown that iron supplementation promotes growth in anaemic 21-day rats¹⁵. Our study extends these findings by demonstrating a comparable weight gain effect in adult, iron-replete rats, thereby providing new evidence that the micro-encapsulated iron formulation LIPOFER™ can enhance body weight gain even when baseline iron stores are adequate. On the other hand, the lower feed efficiency reported in the animals supplemented with FeSO_4 could suggest a reduction in nutrient absorption induced by FeSO_4 supplementation. Similar results were observed in a preliminary study performed in our group comparing FeSO_4 with LIPOFER™ at an equivalent human dosage of 200 mg/day for 15 days (unpublished data). The supplementation with a high dose of FeSO_4 caused a significant reduction in body weight gain and feed efficiency compared to the control group. This result would indicate a decreased conversion of energy intake into body weight gain in this group of animals, which could be interpreted as a possible adverse effect on nutrient absorption. Interestingly, these results were not reported in those rats supplemented with LIPOFER™ at high doses, again suggesting Def-LFe1 supports greater nutrient absorption.

Regarding the capacity of the different iron supplements to reverse iron deficiency, all iron supplements were able to fully recover haemoglobin levels on day 14. In addition, HRE was determined by the ratio of dietary and supplemented iron conversion into circulating haemoglobin. Considering that the significant role of iron in haemoglobin is oxygen binding, the lower HRE obtained at day 21 when compared to day 14 in all Fe-supplemented groups would indicate that haemoglobin has bound all possible oxygen and, consequently, a lower amount of the iron-supplemented during those days was incorporated into haemoglobin. RBV of Fe from the tested iron supplements did not show significant differences among groups. Even though FeSO_4 is considered the reference compound for food fortification due to its high bioavailability, the present study reported a similar bioavailability among the different iron supplementations. A possible explanation for the absence of differences noted in the present study could be the higher doses administered of each compound (6.2 mg Fe/kg body weight, equivalent to a human dose of 80 mg of elemental iron). In other words, the efficacy of all products administered at a high dose in rapidly reversing anaemia within 14 days may account for the absence of differences in the bioavailability among the different iron supplements.

Iron deficiency anaemia is marked by reduced levels of serum iron. Since transferrin is the body's primary iron carrier, iron deficiency leads to diminished transferrin saturation¹⁶. Furthermore, transferrin levels, UIBC, and TIBC are elevated in cases of anaemia, exhibiting an inverse correlation with circulating iron levels. This occurs because the liver produces more transferrin to optimise the utilisation of the limited iron supply¹⁷. In the present study, all iron-supplemented groups displayed a significant increase in serum iron levels and transferrin saturation and a sharp decrease in TIBC, transferrin and UIBC levels. Interestingly, LIPOFER™ induced a more

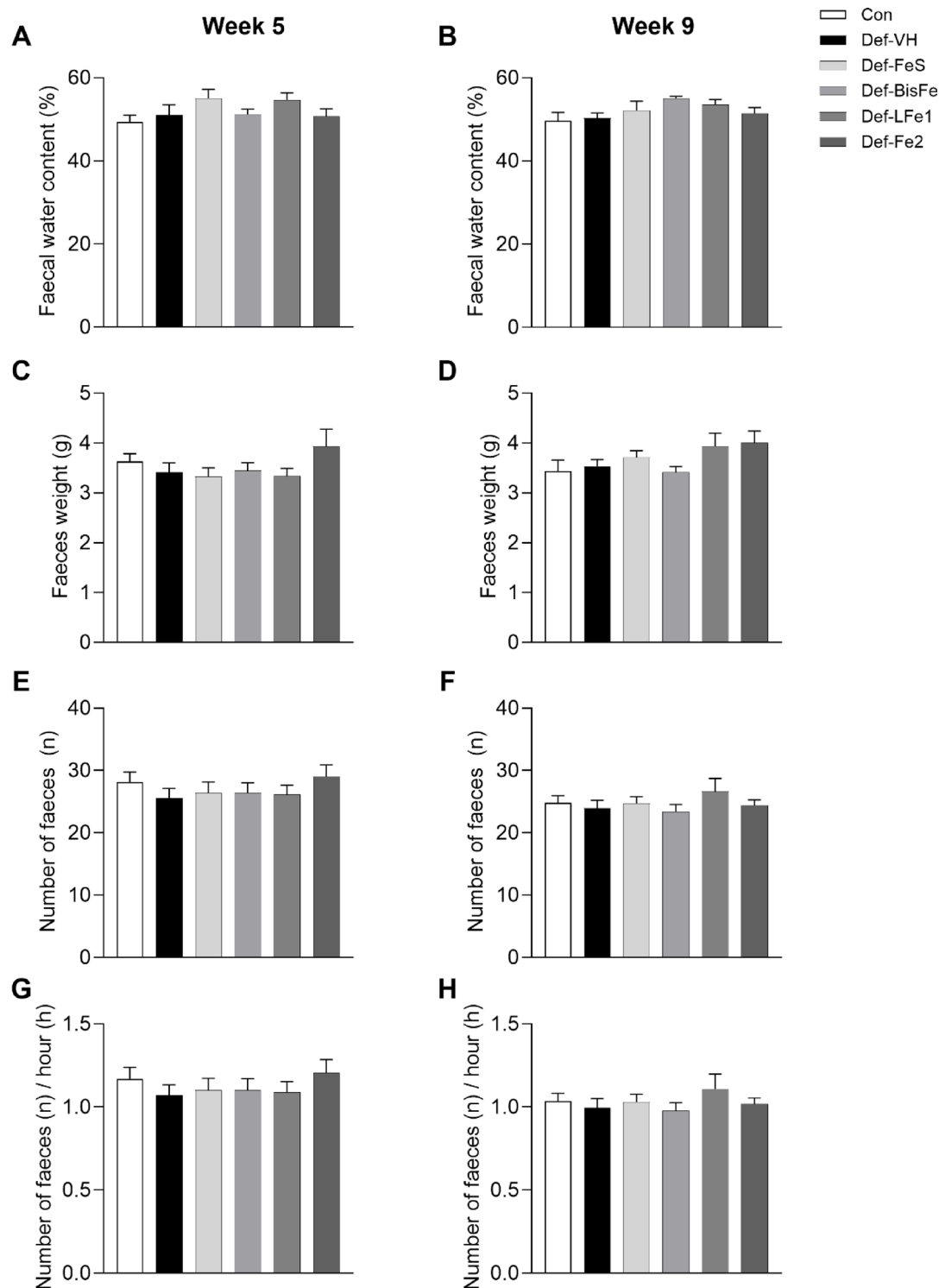


Fig. 9. Faecal water content (%) (A, B), faeces weight (C, D), number of faeces (E, F) and number of faeces/hour (G, H) reported on week 5 and week 9 of the side effects evaluation period. All values are presented as mean $\pm$ SEM.

significant reduction in TIBC and transferrin levels than the iron pyrophosphate microencapsulated 2 (Def-Fe2 group), suggesting a higher efficacy in iron level repletion in this group. Additionally, the higher haemoglobin and lower TIBC levels displayed by Def-LFe1 animals in comparison with Def-Fe2 animals at the end of the study would suggest a better absorption rate and a higher conversion from supplemented iron to circulating haemoglobin.

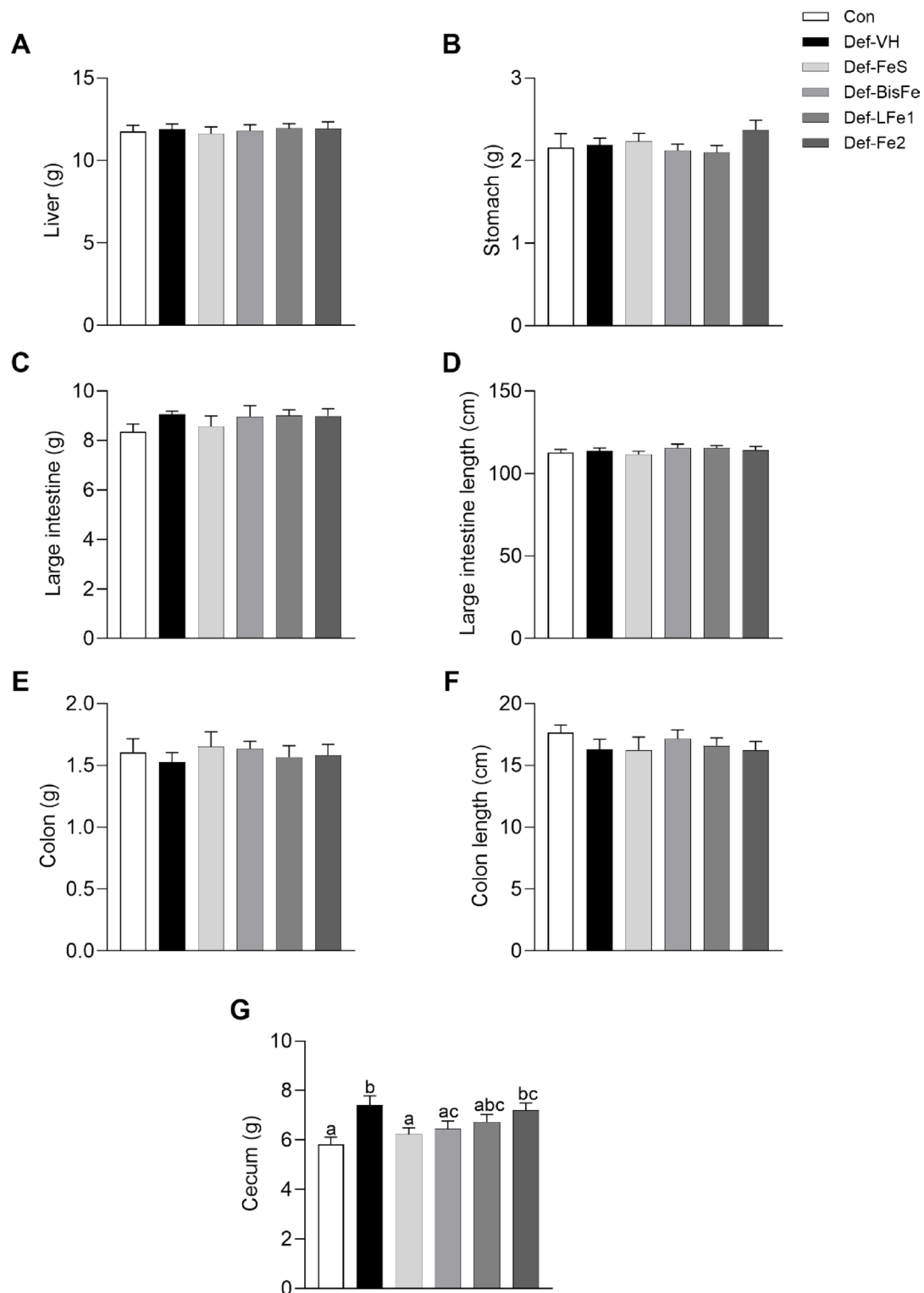


Fig. 10. Liver (A), stomach (B), large intestine (C), colon (E) and cecum (G) weight and large intestine (D) and colon (F) length were measured at the sacrifice of the animals. All values are presented as mean $\pm$ SEM. ^{abc} Mean values with different letters are significantly different among the groups ($p < 0.05$, 1w-ANOVA, Duncan's post hoc test).

The predominant gastrointestinal side effects associated with iron consumption include nausea/vomiting, constipation or diarrhoea, flatulence, and epigastric distress¹⁸. The lack of differences in faecal water content, weight, number and frequency of faecal depositions—valuable markers of gastrointestinal side effects¹⁸ indicated an absence of diarrhea and constipation symptoms in all the experimental groups. In terms of biometric

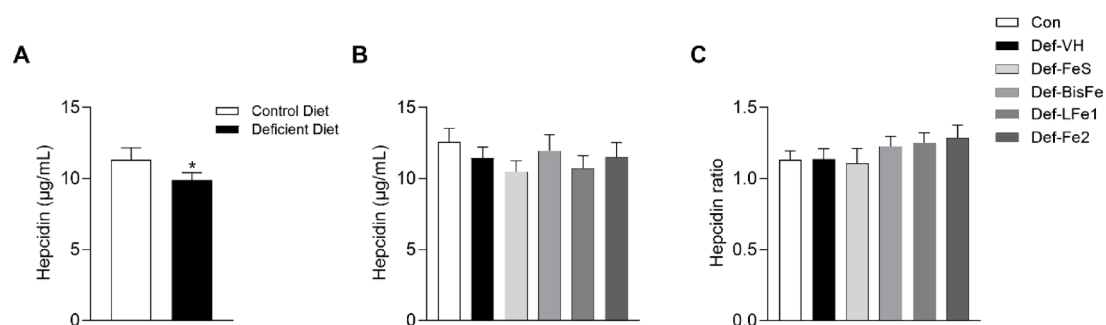


Fig. 11. Hepcidin levels at the beginning of the repletion period in control ($n=10$) and deficient diets-fed animals ($n=50$) (A), hepcidin levels at the end of the study (B) and hepcidin induction ratio (C). All values are presented as mean $\pm$ SEM. * $p < 0.05$ Mann-Whitney test vs. Con group.

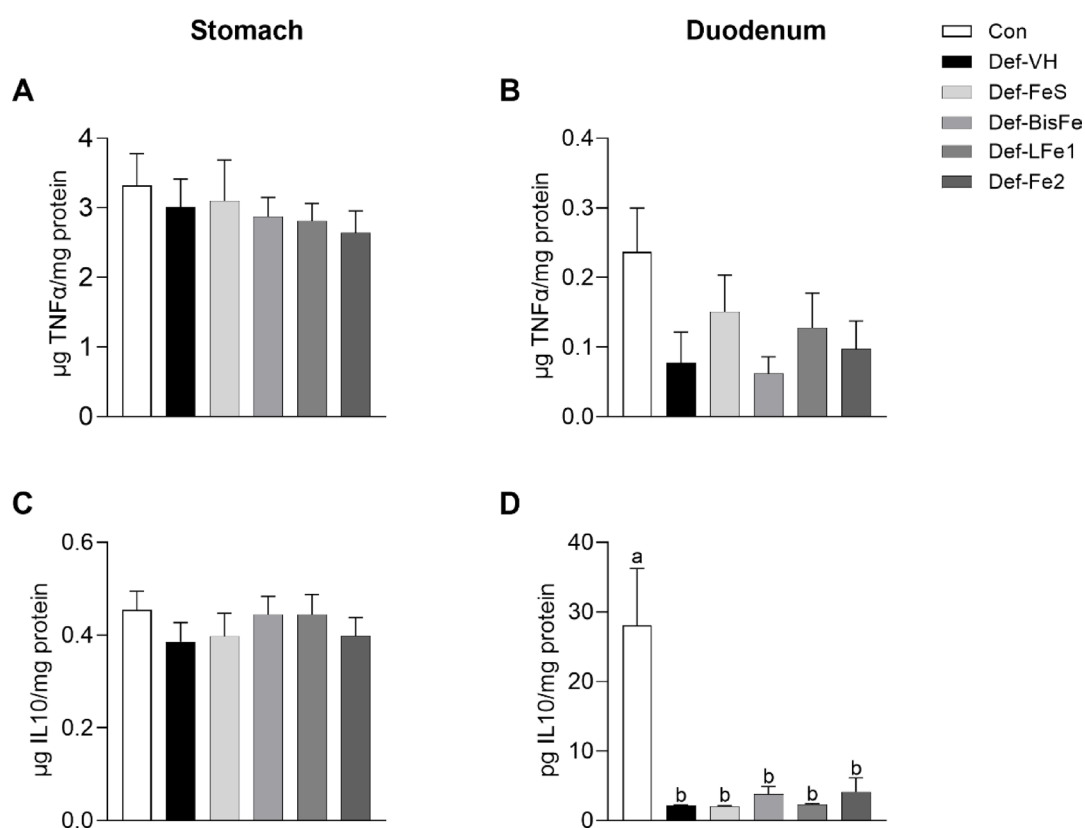


Fig. 12. TNF α levels in the stomach (A) and duodenum (B) and IL-10 levels in the stomach (C) and duodenum (D). All values are presented as mean $\pm$ SEM.

parameters, no differences in the large intestine and colon length were reported among groups. Shortening of the small intestine is recognised as a key characteristic of inflammatory bowel diseases (IBD) such as Crohn's disease and ulcerative colitis^{19,20}. Thus, the absence of structural alterations in these tissues reinforces the lack of side effects induced by the different iron supplementations evaluated in the present study.

Hepcidin, a peptide hormone derived from the liver, is crucial in regulating systemic iron homeostasis. Hepcidin operates by inhibiting the flow of iron into the plasma through several mechanisms, including the absorption in the duodenum, the release from macrophages recycling old red blood cells, and the mobilisation of stored iron from hepatocytes²¹. Two different mechanisms regulate the production of hepcidin: (a) it is increased in response to plasma and liver iron levels, serving as a feedback mechanism to maintain stable body iron levels, and (b) it is increased by inflammation as a defence mechanism to limit extracellular iron availability to microbes²². As it was reported in the present study, it has been described that in conditions of iron deficiency, the reduction of hepcidin production is an adaptation mechanism that facilitates dietary and pharmacological iron absorption²³.

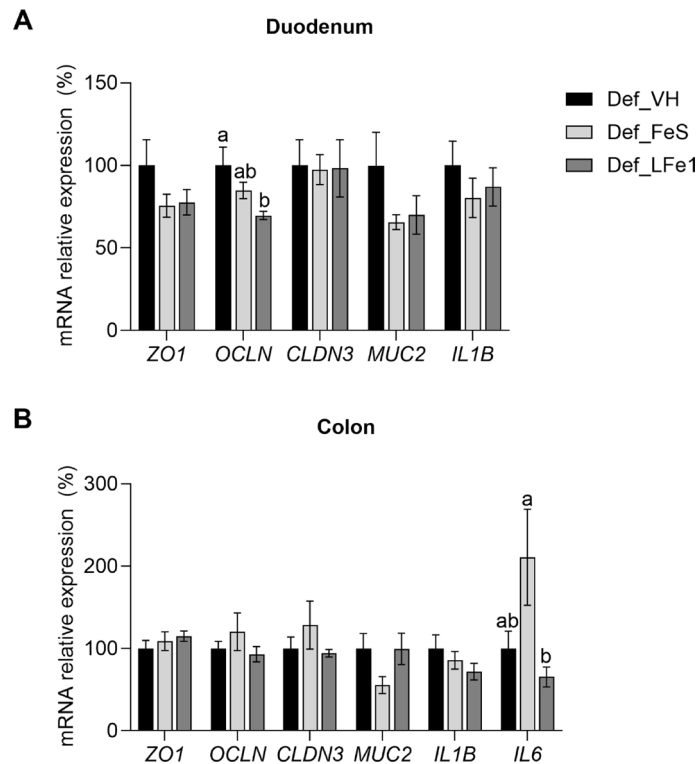


Fig. 13. mRNA expression levels of genes encoding tight junction proteins and inflammation mediators in the duodenum (A) and colon (B) in Def_VH, Def_FeS and Def_LFe1 groups. Genes encoding zonulin 1 (*ZO1*), occludin (*OCLN*), claudin 3 (*CLDN3*), mucin 2 (*MUC2*), interleukin 1 beta (*IL1β*) and interleukin 6 (*IL6*) were determined in the present study. All values are presented as mean $\pm$ SEM. ^{abc} Mean values with different letters are significantly different among the groups ($p < 0.05$, 1w-ANOVA, Duncan's post hoc test).

In the present study, the hepcidin induction ratio was calculated from the plasmatic hepcidin levels measured at the beginning of the repletion period and those obtained at the end of the study. All the experimental groups of the study showed an increase in hepcidin levels at the end of the study, although no differences were reported among the Fe-treated groups.

It is widely described that iron supplementation stimulates the production of hepcidin, which limits iron absorption obtained from the diet and further regulates iron storage as a mechanism to avoid iron overload²². However, as mentioned above, increased hepcidin levels have also been associated with increased inflammation, as pro-inflammatory cytokines also regulate its production²¹. Experimental evidence suggests that excessive iron in the lumen may also harm the intestinal mucous membrane. Despite its key role in cellular processes, free iron in the intestine can generate toxic free radicals that may directly impair intestinal epithelium's integrity via oxidative stress. The production of reactive oxygen species (ROS), in turn, can trigger an inflammatory response both locally in the gut and at a systemic level²⁴. Consequently, inflammatory biomarkers were quantified in stomach and duodenal samples from all experimental groups to gain insight into this issue.

The lack of differences among iron-supplemented animals regarding inflammatory biomarkers at the level of stomach and duodenum would indicate that the chronic administration of an equivalent human dose of 80 mg of elemental iron through different iron supplements is not promoting a pro-inflammatory response in the animals in these specific areas. These facts are corroborated by the absence of differences in gut measurements (large intestine and colon weight and length), faecal characteristics assessment (faecal consistency and humidity and bowel movements' frequency) and in the quantification of circulating hepcidin levels.

Considering the suggested impact on nutrient absorption induced by chronic supplementation of FeSO_4 , the analysis of the expression of genes associated with intestinal inflammation and permeability in duodenum and colon was suggested to provide mechanistic information of the subtle effects induced by FeSO_4 and LIPOFER™ supplements. Claudin and occludin are considered the most important tight junction membrane proteins, which determine the permeability of the intestinal barrier^{25,26}. Related to them, zonulin 1 is also considered an essential regulator of tight junction assembly in epithelia²⁶. A decrease in the protein levels of these tight junctions would lead to an increased permeability in the gut²⁷ resulting in a state of poor nutrient assimilation²⁸. Finally, the gene encoding the mucin 2 protein (*MUC2*) is involved in the production of mucin, one of the main components in the formation of the mucous layer of the intestinal barrier, which protects the intestine from external aggressions²⁹. Given the lack of reported intestinal disturbances in the present study, the lower mRNA levels of *OCLN* observed in Def_LFe1 animals did not appear to be relevant. A possible explanation for these results is that gene expression data do not always correlate with protein levels. Therefore, quantifying the protein levels of some of these tight junctions would help clarify this issue.

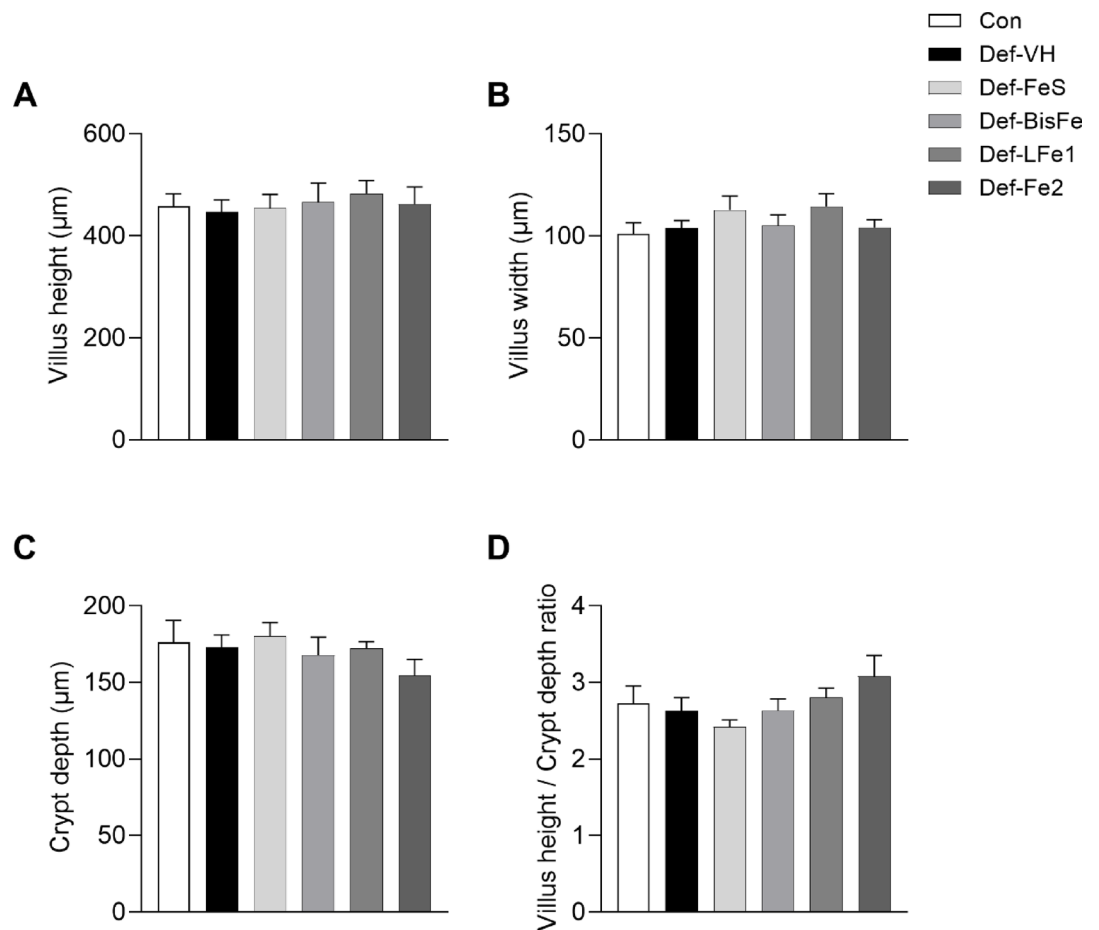


Fig. 14. Villus height (A), width (B), crypt depth (C) and villus height to crypt depth ratio (D) in duodenal samples. All values are presented as mean ± SEM.

In the samples obtained from the colon, Def-LFe1 animals displayed higher *MUC2* mRNA levels and lower *IL6* mRNA than those animals treated with FeSO_4 . *IL-6* plays a central role in the pathogenesis of inflammatory-based disorders such as inflammatory bowel disease³⁰. In addition, previous studies have reported that iron overloading induced by the chronic supplementation of iron can promote intestinal inflammation (increased *IL-6*) and alter the intestinal barrier (decreased *Muc-2*)³¹. More specifically, rodent studies have shown that FeSO_4 administration led to increased levels of inflammation-related biomarkers, such as *IL6* and *TNFα*^{32,33}. These data come from colitis models, where pre-existing mucosal injury and heightened oxidative stress make these animals especially sensitive to the pro-oxidant properties of soluble ferrous salts. In the present study, the obtained results in the colon could suggest a slight stimulation of pro-inflammatory biomarkers production by the chronic FeSO_4 supplementation and, consequently, a subtle negative effect on the intestinal mucosal barrier in this tissue, a result that was not reported in Def-LFe1 animals. However, as previously stated, these results should be validated through a protein expression analysis of these biomarkers in the colonic tissue.

Prior research has indicated that extended and high-dose iron supplementation regimens can harm the intestinal mucosa, primarily triggered by oxidative stress and inflammatory mechanisms³⁴. In this study, histological examinations on the duodenal tissue and stomach were conducted to investigate the potential adverse impacts of iron supplementation. Interestingly, Def-LFe1 animals exhibited a significantly higher villus height to crypt depth ratio than Def-FeS animals. Villus height to crypt depth ratio is considered one of the most significant parameters associated with nutrient absorption. An increase in this ratio is correlated with improved digestion and absorption of nutrients³⁵. Interestingly, these results would be aligned with the higher body weight gain and feed efficiency differences reported in Def-LFe1 groups compared to Def-FeS animals, suggesting a reduced nutrient absorption caused by chronic supplementation of FeSO_4 in comparison with those supplemented with LIPOFER™. However, further studies should be conducted to provide more insight into this issue.

The quantification of iron foci in the duodenal tissue showed a residual increase in the amount of iron foci in Def-LFe1 animals than in Def-FeS animals and the Def-Fe2 group, the latter also receiving microencapsulated iron pyrophosphate. As shown in the representative images of Prussian blue iron staining, iron foci were detected and quantified in the lamina propria of the intestinal mucosa, precisely at the top of the villi. Lamina propria is characterised by containing several immune cells, such as macrophages, eosinophils, and lymphocytes³⁶.

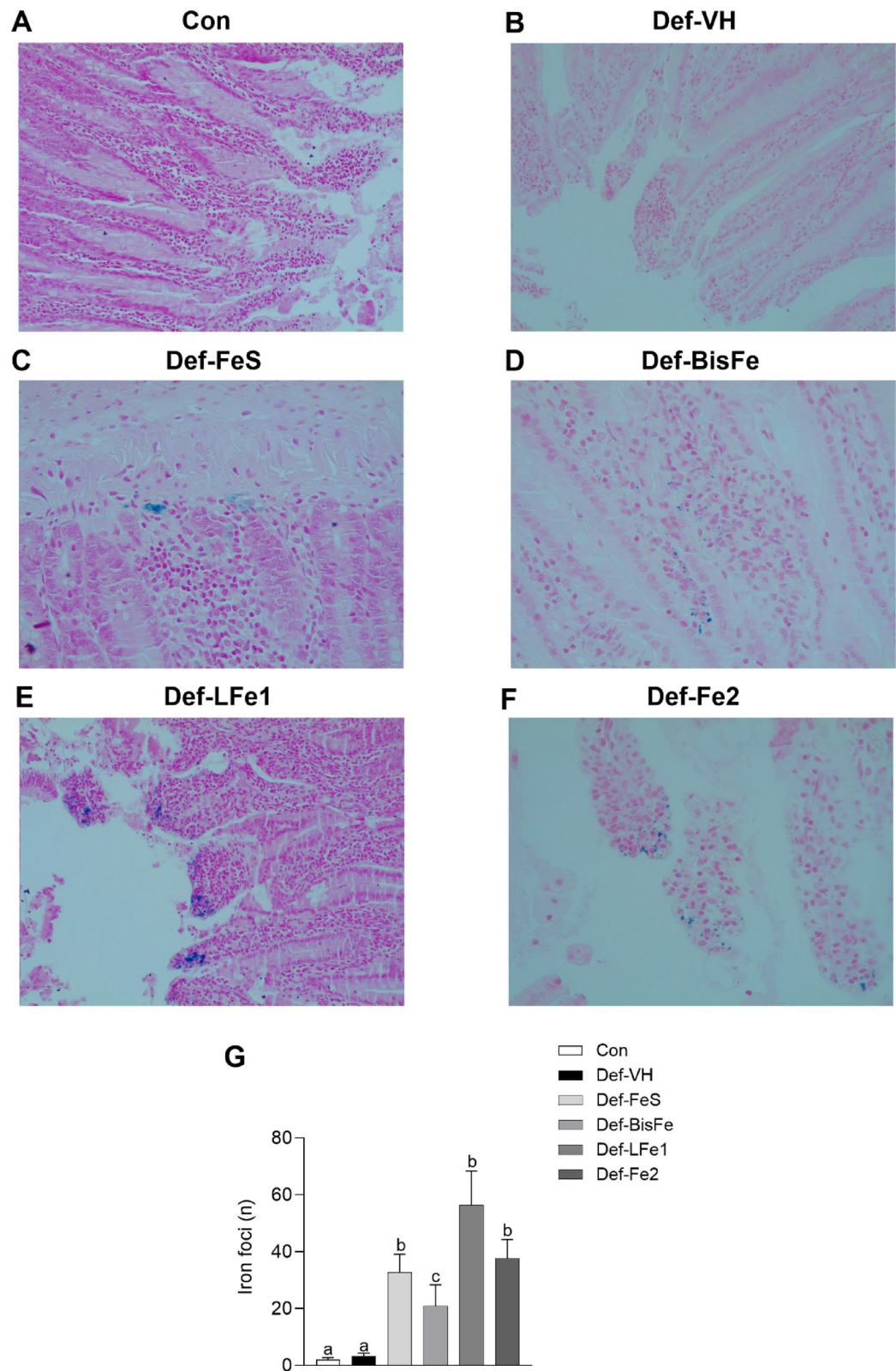


Fig. 15. Representative images of Prussian blue iron staining in duodenal samples in all the experimental groups (A–F) and number of iron foci in 4 different duodenal sections (G). All values are presented as mean $\pm$ SEM. ^{abc} Mean values with different letters are significantly different among the groups ($p < 0.05$, 1w ANOVA, Duncan's post hoc test).

Macrophages play a pivotal role in iron trafficking by recycling the substantial amount of iron acquired through haemoglobin degradation, demonstrating a unique adaptability to manage and store iron quantities that might be toxic to other cell types³⁷. Interestingly, duodenal macrophages have also been described to modulate iron absorption and transfer to circulation by controlling transferrin availability in the lamina propria villi³⁸. Although additional studies should be performed to shed more light on the relevance of these findings, a potentially higher iron recycling rate performed by macrophages could account for the higher number of iron foci quantified in the group supplemented with LIPOFER™ microcapsules compared to the groups supplemented with FeSO₄ and the other microencapsulated ferric pyrophosphate product.

Previous *in vitro* studies have proposed an alternative mechanism involving the absorption of iron microparticles^{39,40}. Microfold (M) cells represent a distinct subset of intestinal epithelial cells found in the gut-associated lymphoid tissue (GALT) of the Peyer's patches in the small intestine. These cells are recognised for triggering mucosal immune responses on the apical membrane of M cells, facilitating the transportation of microbes and insoluble particles across the epithelial cell layer from the gut lumen to the lamina propria. This transfer enables interactions with immune cells in the lamina propria^{41,42}. Despite the well-described role of these cells in promoting the exposure of microbial antigens to the mucosal immune system, it has been suggested that M-cells can also facilitate the intestinal absorption of micronised iron particles⁴³. Thus, the higher number of iron foci observed in Def-LFe1 animals could be attributed to enhanced iron absorption through this alternative pathway by this product.

Conclusions

Considering the study's primary results, it can be concluded that each iron source fully reverted iron deficiencies after 14 days of supplementation. However, based on the decrease in TIBC and transferrin levels, coupled with the higher haemoglobin levels, LIPOFER™ microcapsules demonstrated a higher absorption rate than the Sunactive® product, further highlighting the efficacy of this microencapsulation technology. Similarly, LIPOFER™ supplementation resulted in a higher number of iron foci, further emphasising its enhanced absorption through M-cells compared to the Ferrous bisglycinate.

Interestingly, Def-LFe1 supplementation also resulted in a greater feed efficiency than the other iron supplements. Regarding potential side effects, FeSO₄-treated animals promoted subtle increases in markers of intestinal inflammation and altered intestinal barrier parameters, effects that were not detected in the other iron-supplemented groups, including Def-LFe1. This study has highlighted the ability of the LIPOFER™ technology to be an effective product for reversing iron deficiency while minimising any unwanted side effects frequently reported for other iron supplements.

Data availability

The data presented in this study are available on request from the corresponding author.

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Author contributions

Conceptualisation, R.M.-C. and A.C. (Antoni Caimari); Methodology, R.M.-C., Y.T., A.A., and J.M.A.; Investigation, R.M.-C., Y.T., A.A., J.M.A. and A.C. (Antoni Caimari); Data curation, R.M.-C., Y.T. and J.M.A.; Writing—original draft preparation, R.M.-C.; Writing—review and editing, R.M.-C., A.C. (Antoni Caimari); Supervision, R.M.-C., Y.T. and J.M.A.; Project administration, R.M.-C., Y.T. and J.M.A.; Visualization, R.M.-C., Y.T., A.A., T.N., A.C. (Alan Connolly), J.M.A. and A.C. (Antoni Caimari). All authors have read and agreed to the published version of the manuscript.

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Declarations

Competing interests

The authors declare no competing interests.

Institutional review board statement

The Animal Ethics Committee of the UTNS-REUS (Reus, Spain) and the Generalitat de Catalunya approved all the procedures. The study was carried out following the Guide for the Care and Use of Laboratory Animals and in accordance with the European provisions of Directive 86/609/EEC on the protection of animals used for experimental and other scientific purposes.

Additional information

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